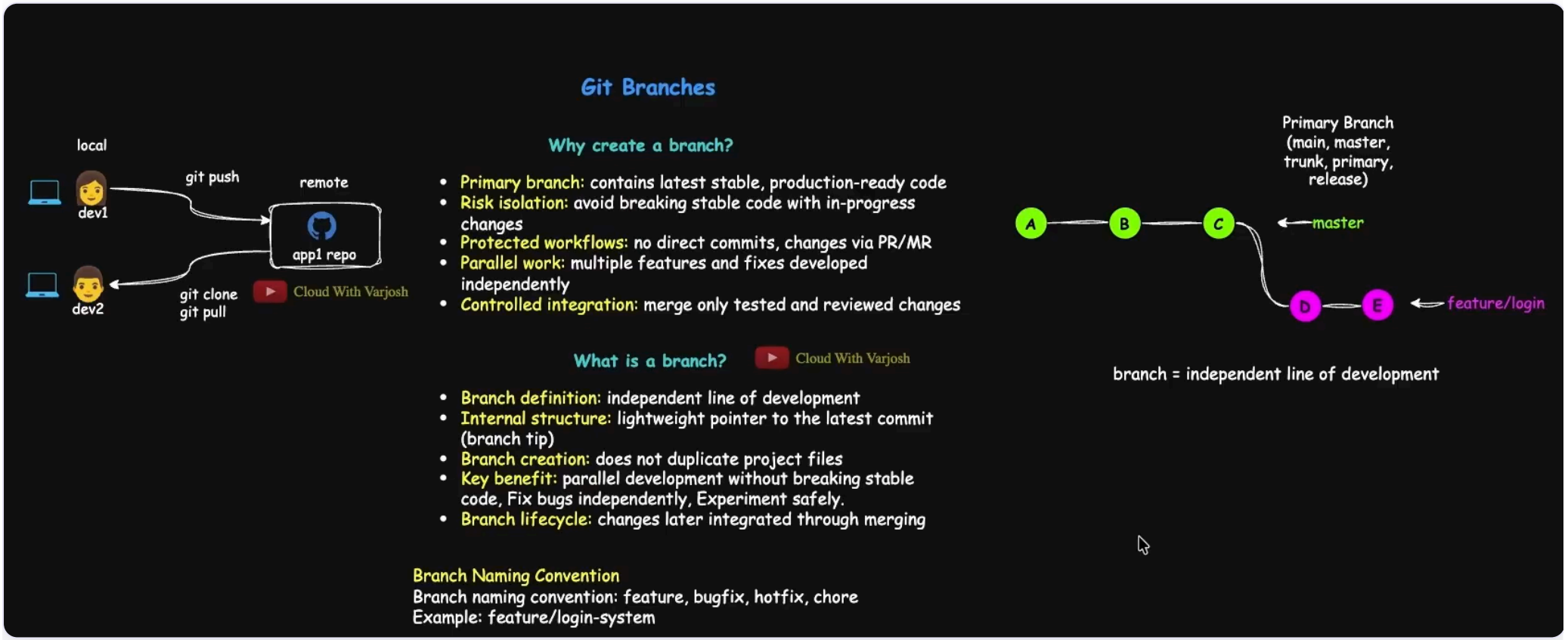
GIT TRACK · SCREENSHOTS

# Chapter 02 — Git Branches & Merge Pointers, HEAD, Fast-forward & Three-way Merge

Git Masterclass · Part 1 · stills from the session, in order · 2026-10-13

# 45 screenshots


Git Branches
local
dev1
git push
remote
app1 repo
dev2
git clone
git pull
Cloud With Varjosh
Why create a branch?
Primary branch: contains latest stable, production-ready code
Risk isolation: avoid breaking stable code with in-progress changes
Protected workflows: no direct commits, changes via PR/MR
Parallel work: multiple features and fixes developed independently
Controlled integration: merge only tested and reviewed changes
What is a branch?
Cloud With Varjosh
Branch definition: independent line of development
Internal structure: lightweight pointer to the latest commit (branch tip)
Branch creation: does not duplicate project files
Key benefit: parallel development without breaking stable code, Fix bugs independently, Experiment safely.
Branch lifecycle: changes later integrated through merging
Branch Naming Convention
Branch naming convention: feature, bugfix, hotfix, chore
Example: feature/login-system
Primary Branch (main, master, trunk, primary, release)
A
B
C
master
D
E
feature/login
branch = independent line of development

## Common Branch Naming Conventions

While Git does not enforce any naming rules, teams usually follow conventions to make branch purpose immediately clear.

Common branch prefixes include:

- **`feature/`** → new functionality (e.g., `feature/login-system`, `feature/terraform-vpc-setup`)
- **`bugfix/`** → non-critical bug fixes (e.g., `bugfix/payment-calculation-error`, `bugfix/helm-values-misconfig`)
- **`hotfix/`** → urgent production fixes (e.g., `hotfix/login-crash`, `hotfix/eks-nodegroup-failure`)
- **`release/`** → preparing a production release (e.g., `release/v1.2.0`, `release/k8s-manifests-v2`)
- **`experiment/`** or **`spike/`** → short-lived experiments or investigations (e.g., `experiment/graphql-adoption`, `spike/argo-rollout evaluation`)
- **`chore/`** → maintenance tasks such as dependency upgrades (e.g., `chore/update-spring-boot-version`, `chore/upgrade-terraf modules`)

These conventions help teams quickly understand the **intent and lifecycle of a branch**.

Once work on a branch is complete, the changes can later be **merged back into the primary branch**. We will discuss **what merging m and how it works shortly**.

# Atomic Development in Git

In Git, both **commits** and **branches** are most effective when they represent **small, focused units of work.**

This principle is known as **atomic development.**

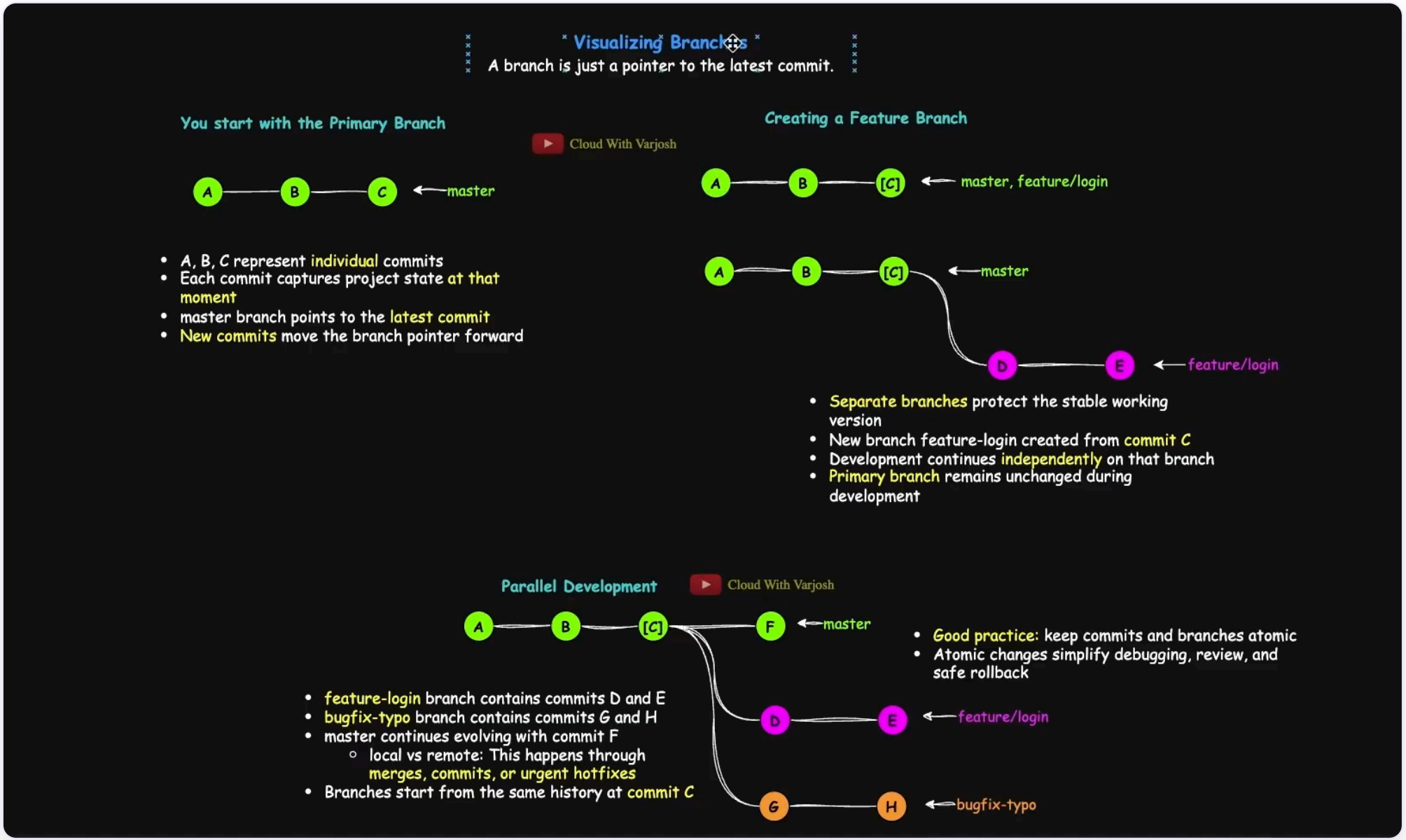
An **atomic change** is a **single, self-contained, and logically complete unit of work** that can be:

- understood independently
- reviewed independently
- tested independently
- reverted independently

Atomicity applies at two levels:

- **Atomic commits** → each commit represents one logical change
- **Atomic branches** → each branch represents one logical task

**Key idea:** One branch, one purpose. One commit, one logical change.


Visualizing Branches
A branch is just a pointer to the latest commit.
You start with the Primary Branch
Cloud With Varjosh
A
B
C
master
A, B, C represent individual commits
Each commit captures project state at that moment
master branch points to the latest commit
New commits move the branch pointer forward
Creating a Feature Branch
A
B
[C]
master, feature/login
A
B
[C]
master
D
E
feature/login
Separate branches protect the stable working version
New branch feature-login created from commit C
Development continues independently on that branch
Primary branch remains unchanged during development
Parallel Development
Cloud With Varjosh
A
B
[C]
F
master
D
E
feature/login
G
H
bugfix-typo
feature-login branch contains commits D and E
bugfix-typo branch contains commits G and H
master continues evolving with commit F
local vs remote: This happens through merges, commits, or urgent hotfixes
Branches start from the same history at commit C
Good practice: keep commits and branches atomic
Atomic changes simplify debugging, review, and safe rollback

## git init

```
root@cka-w2 ~/project-files
$ git init
hint: Using 'master' as the name for the initial branch. This default branch na
me
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /root/project-files/.git/
root@cka-w2 ~/project-files
$
```

**07** / 45 chore: initialize application (0e3ffcd)

```
root@cka-w2 ~/project-files
$ echo "application initialized" > app.txt
root@cka-w2 ~/project-files
$ git add app.txt
root@cka-w2 ~/project-files
$ git commit -m "chore: initialize application"
[master (root-commit) 0e3ffcd] chore: initialize application
 1 file changed, 1 insertion(+)
 create mode 100644 app.txt
root@cka-w2 ~/project-files (master)
$
```

 Commit without git add → no changes added

```
$  echo "updated aplication header styling" >> app.txt
root@cka-w2 ~/project-files (master)
$ git commit -m "style: improve application header styling"
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app.txt

no changes added to commit (use "git add" and/or "git commit -a")
root@cka-w2 ~/project-files (master)
$ git add .
root@cka-w2 ~/project-files (master)
$ git commit -m "style: improve application header styling"
[master fcaef20] style: improve application header styling
 1 file changed, 1 insertion(+)
root@cka-w2 ~/project-files (master)
$
```

feat: add footer section

```
root@cka-w2 ~/project-files (master)
$ git log --oneline
fcaef20 (HEAD -> master) style: improve application header styli
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (master)
$ echo "Added footer section to application" >> app.txt
root@cka-w2 ~/project-files (master)
$ git add .
root@cka-w2 ~/project-files (master)
$ git commit -m "feat: add footer section"
[master 9aaeab0] feat: add footer section
 1 file changed, 1 insertion(+)
root@cka-w2 ~/project-files (master)
$
```

git log --oneline — 3 commits

```
root@cka-w2 ~/project-files (master)
$ git log --oneline
9aaeab0 (HEAD -> master) feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (master)
$
```

cat app.txt

```
root@cka-w2 ~/project-files (master)
$ cat app.txt
application initialized
updated aplication header styling
Added footer section to application
root@cka-w2 ~/project-files (master)
$
```

## Full git log

```
root@cka-w2 ~/project-files (master)
$ git log
commit 9aaeab01f25cf5a14ed26f13d7dcb8f9f8df874e (HEAD -> master)
Author: vishal <vishalbhilare@test.com>
Date:   Tue Oct 13 09:50:30 2026 +0000

    feat: add footer section

commit fcaef207c1bf21bcea40334cbd18a806dbc1edac
Author: vishal <vishalbhilare@test.com>
Date:   Tue Oct 13 09:47:04 2026 +0000

    style: improve application header styling

commit 0e3ffcd52e9e990df3d508bc3bd40c63890cbf99
Author: vishal <vishalbhilare@test.com>
Date:   Tue Oct 13 09:43:29 2026 +0000

    chore: initialize application
root@cka-w2 ~/project-files (master)
```

## git log --oneline

```
root@cka-w2 ~/project-files (master)
$ git log --oneline
9aaeab0 (HEAD -> master) feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (master)
$
```

git branch feature/login + git checkout

```
root@cka-w2 ~/project-files (master)
$ git branch feature/login
root@cka-w2 ~/project-files (master)
$ git checkout feature/login
Switched to branch 'feature/login'
root@cka-w2 ~/project-files (feature/login)
$
```

15 / 45 git branch — * current

```
root@cka-w2 ~/project-files (feature/login)
$ git branch
* feature/login
  master
```

16 / 45 Cannot delete the current branch

```
root@cka-w2 ~/project-files (feature/login)
$ git branch -d feature/login
error: cannot delete branch 'feature/login' used by worktree at '/root/project-
files'
root@cka-w2 ~/project-files (feature/login)
$
```

17 / 45 Switch to master, then git branch -d

```
$ git checkout master
Switched to branch 'master'
root@cka-w2 ~/project-files (master)
$ git branch -d feature/login
Deleted branch feature/login (was 9aaeab0).
root@cka-w2 ~/project-files (master)
$
```

git checkout -b feature/login



Switched to a new branch

```
root@cka-w2 ~/project-files (master)
$ git checkout -b feature/login
Switched to a new branch 'feature/login'
root@cka-w2 ~/project-files (feature/login)
$ git branch
* feature/login
  master
root@cka-w2 ~/project-files (feature/login)
$
```

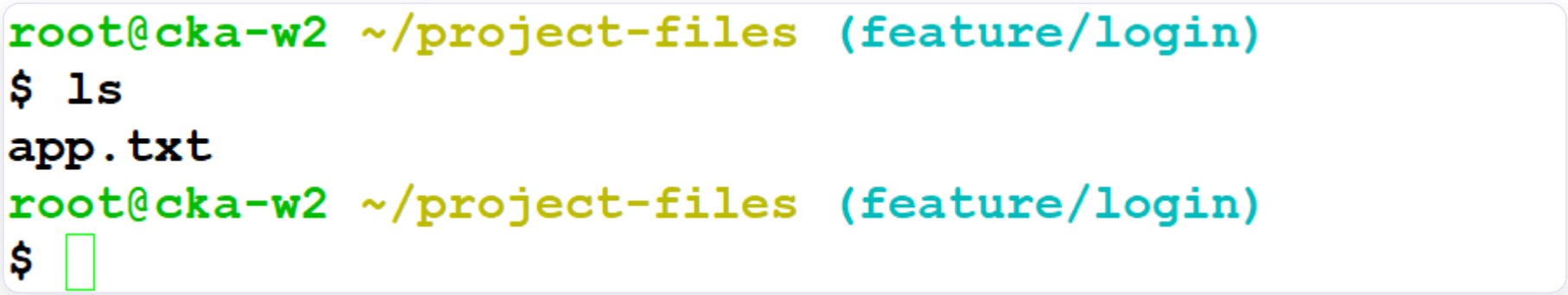
20 / 45 HEAD → feature/login, master

```
root@cka-w2 ~/project-files (feature/login)
$ git log --oneline
9aaeab0 (HEAD -> feature/login, master) feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (feature/login)
$
```

## Switching branches — only HEAD moves

```
root@cka-w2 ~/project-files (feature/login)
$ git checkout master
Switched to branch 'master'
root@cka-w2 ~/project-files (master)
$ git log --oneline
9aaeab0 (HEAD -> master, feature/login) feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (master)
$ git checkout feature/login
Switched to branch 'feature/login'
root@cka-w2 ~/project-files (feature/login)
$ git log --oneline
9aaeab0 (HEAD -> feature/login, master) feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (feature/login)
$
```

22 / 45 feature/login sees app.txt

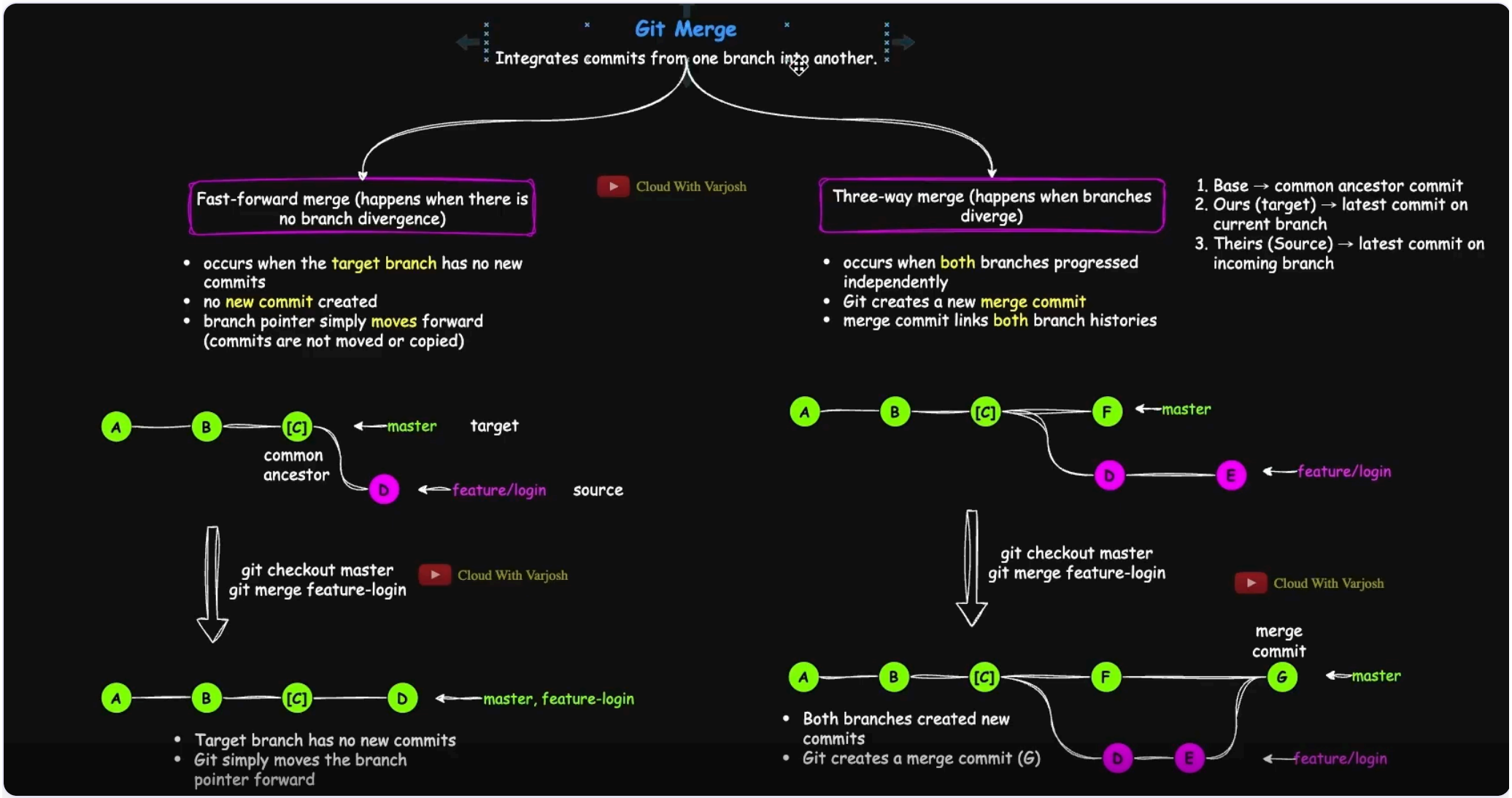


```
root@cka-w2 ~/project-files (feature/login)
$ echo "Login feature implementation started" > login .txt
root@cka-w2 ~/project-files (feature/login)
$ git add .
root@cka-w2 ~/project-files (feature/login)
$ git commit -m "feat: implement login functionality"
[feature/login 13b779a] feat: implement login functionality
 1 file changed, 1 insertion(+)
 create mode 100644 login
root@cka-w2 ~/project-files (feature/login)
$ git log --oneline
13b779a (HEAD -> feature/login) feat: implement login functionality
9aaeab0 (master) feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (feature/login)
$
```

login exists only on feature/login

```
root@cka-w2 ~/project-files (feature/login)
$ ls
app.txt  login
root@cka-w2 ~/project-files (feature/login)
$ cat login
Login feature implementation started .txt
root@cka-w2 ~/project-files (feature/login)
$ git checkout master
Switched to branch 'master'
root@cka-w2 ~/project-files (master)
$ ls
app.txt
root@cka-w2 ~/project-files (master)
$
```

# Git merge — fast-forward vs three-way



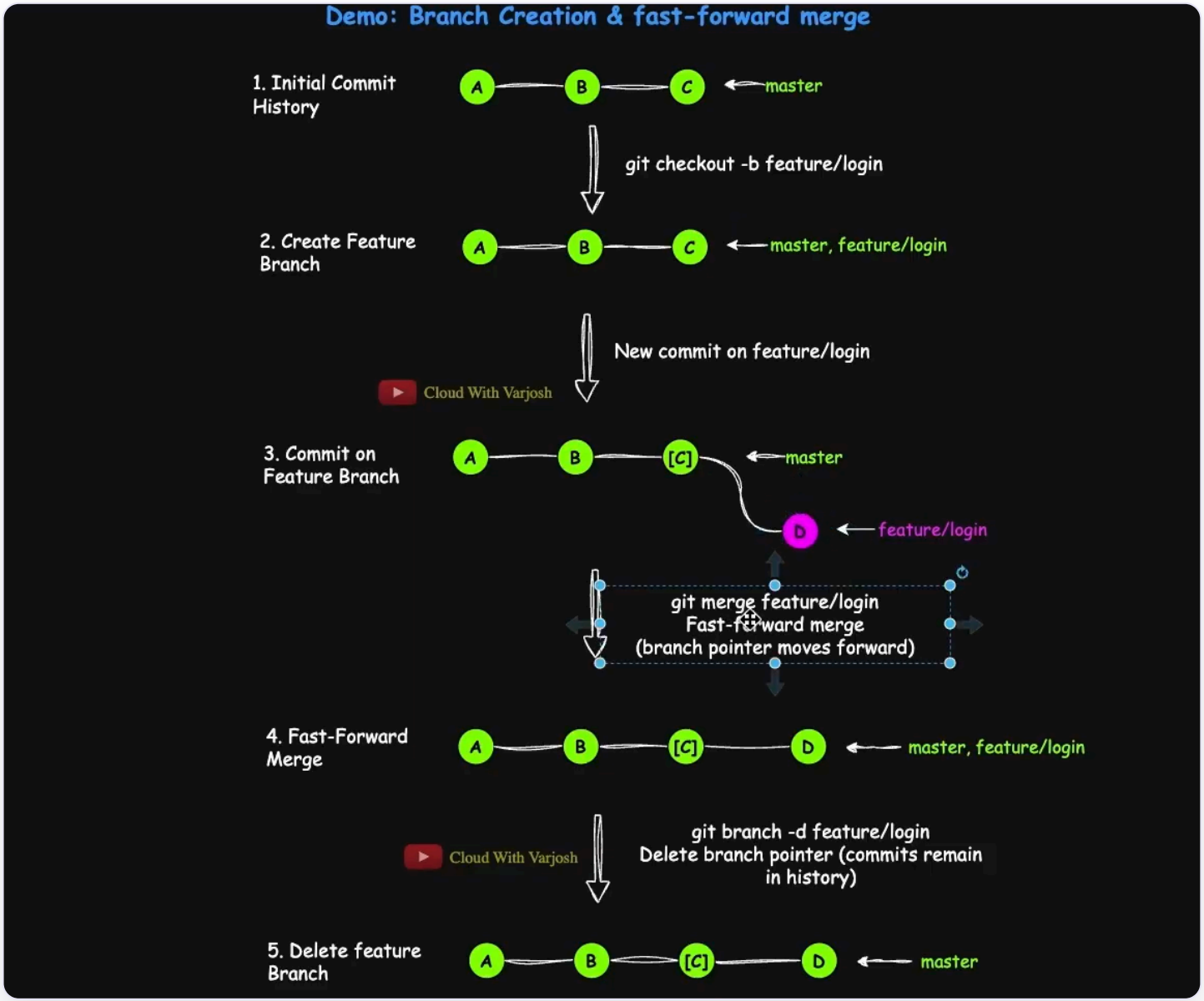
**26** / 45 git merge feature/login → Fast-forward

```
root@cka-w2 ~/project-files (master)
$ git merge feature/login
Updating 9aaeab0..13b779a
Fast-forward
 login | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 login
root@cka-w2 ~/project-files (master)
$ git log --oneline
13b779a (HEAD -> master, feature/login) feat: implement login functionality
9aaeab0 feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (master)
$
```

git reflog — merge feature/login: Fast-forward

```
root@cka-w2 ~/project-files (master)
$ git reflog
13b779a (HEAD -> master, feature/login) HEAD@{0}: merge feature/login: Fast-for
ward
9aaeab0 HEAD@{1}: checkout: moving from feature/login to master
13b779a (HEAD -> master, feature/login) HEAD@{2}: commit: feat: implement logir
 functionality
9aaeab0 HEAD@{3}: checkout: moving from master to feature/login
9aaeab0 HEAD@{4}: checkout: moving from feature/login to master
9aaeab0 HEAD@{5}: checkout: moving from master to feature/login
9aaeab0 HEAD@{6}: checkout: moving from feature/login to master
9aaeab0 HEAD@{7}: checkout: moving from master to feature/login
9aaeab0 HEAD@{8}: commit: feat: add footer section
fcaef20 HEAD@{9}: commit: style: improve application header styling
0e3ffcd HEAD@{10}: commit (initial): chore: initialize application
root@cka-w2 ~/project-files (master)
$
```

# Demo: branch creation & fast-forward merge



git checkout -b bugfix/typo from the tip

```
root@cka-w2 ~/project-files (master)
$ git checkout -b bugfix/typo
Switched to a new branch 'bugfix/typo'
root@cka-w2 ~/project-files (bugfix/typo)
$ ls
app.txt  login
root@cka-w2 ~/project-files (bugfix/typo)
$
```

30 / 45 log --oneline on bugfix/typo

```
$ git log --oneline
13b779a (HEAD -> bugfix/typo, master, feature/login) feat: implement login func
tionality
9aaeab0 feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (bugfix/typo)
$
```

git checkout -b bugfix/typo 0e3ffcd

```
root@cka-w2 ~/project-files (master)
$ git checkout -b bugfix/typo 0e3ffcd
Switched to a new branch 'bugfix/typo'
root@cka-w2 ~/project-files (bugfix/typo)
$
```

Only the first commit in history

```
$ git log --oneline
0e3ffcd (HEAD -> bugfix/typo) chore: initialize application
root@cka-w2 ~/project-files (bugfix/typo)
$
```

app.txt has one line

```
$ git log --oneline
0e3ffcd (HEAD -> bugfix/typo) chore: initialize application
root@cka-w2 ~/project-files (bugfix/typo)
$ ls
app.txt
root@cka-w2 ~/project-files (bugfix/typo)
$ cat app.txt
application initialized
root@cka-w2 ~/project-files (bugfix/typo)
$
```

Delete and recreate bugfix/typo from master

```
Switched to a new branch 'bugfix/typo'
root@cka-w2 ~/project-files (bugfix/typo)
$ git log --oneline
0e3ffcd (HEAD -> bugfix/typo) chore: initialize application
root@cka-w2 ~/project-files (bugfix/typo)
$ ls
app.txt
root@cka-w2 ~/project-files (bugfix/typo)
$ cat app.txt
application initialized
root@cka-w2 ~/project-files (bugfix/typo)
$ git checkout master
Switched to branch 'master'
root@cka-w2 ~/project-files (master)
$ git branch -d bugfix/typo
Deleted branch bugfix/typo (was 0e3ffcd).
root@cka-w2 ~/project-files (master)
$ git checkout -b bugfix/typo
Switched to a new branch 'bugfix/typo'
root@cka-w2 ~/project-files (bugfix/typo)
$
```

35 / 45 All three branches at 13b779a

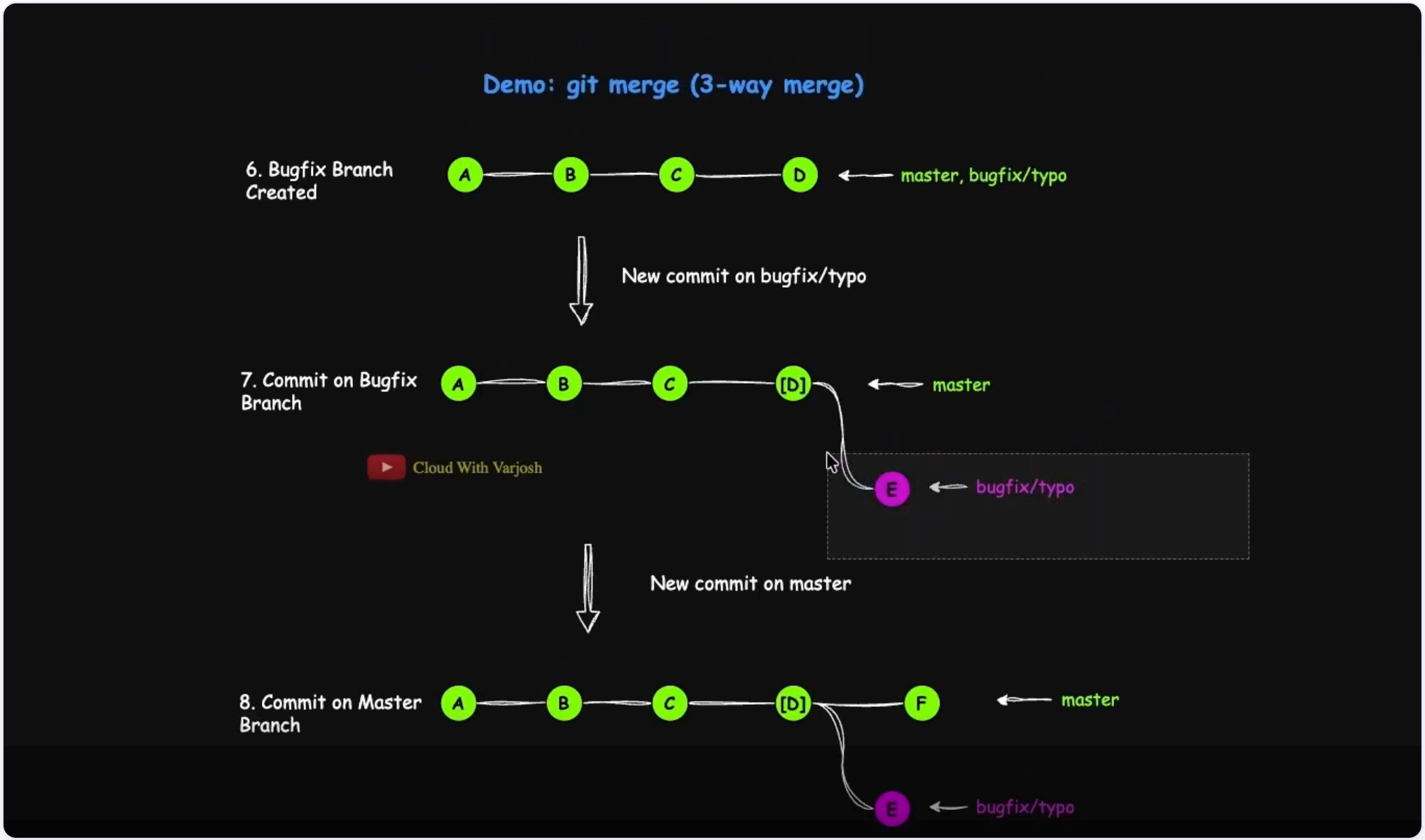
```
$ git log --oneline
13b779a (HEAD -> bugfix/typo, master, feature/login) feat: implement login func
tionality
9aaeab0 feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (bugfix/typo)
$
```

fix: correct footer typo (3fc7beb)

```
root@cka-w2 ~/project-files (bugfix/typo)
$ echo "corrected typo in footer" > bufix.txt
root@cka-w2 ~/project-files (bugfix/typo)
$ git add .
root@cka-w2 ~/project-files (bugfix/typo)
$ git commit -m "fix: correct footer typo"
[bugfix/typo 3fc7beb] fix: correct footer typo
 1 file changed, 1 insertion(+)
 create mode 100644 bufix.txt
root@cka-w2 ~/project-files (bugfix/typo)
$
```

37 / 45 bugfix/typo one commit ahead

```
root@cka-w2 ~/project-files (bugfix/typo)
$  git log --oneline
3fc7beb (HEAD -> bugfix/typo) fix: correct footer typo
13b779a (master, feature/login) feat: implement login functionality
9aaeab0 feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (bugfix/typo)
$
```


Demo: git merge (3-way merge)
6. Bugfix Branch Created
A
B
C
D
master, bugfix/typo
New commit on bugfix/typo
7. Commit on Bugfix Branch
A
B
C
[D]
master
Cloud With Varjosh
E
bugfix/typo
New commit on master
8. Commit on Master Branch
A
B
C
[D]
F
master
E
bugfix/typo

style: improve footer spacing on master (d40c83f)

```
$ git checkout master
Switched to branch 'master'
root@cka-w2 ~/project-files (master)
$ ls
app.txt  login
root@cka-w2 ~/project-files (master)
$ echo "Improved footer spacing" > layout.txt
root@cka-w2 ~/project-files (master)
$ git add .
root@cka-w2 ~/project-files (master)
$ git commit -m "style: improve footer spacing"
[master d40c83f] style: improve footer spacing
 1 file changed, 1 insertion(+)
 create mode 100644 layout.txt
root@cka-w2 ~/project-files (master)
$
```

## Different new files per branch

```
root@cka-w2 ~/project-files (master)
$ ls
app.txt  layout.txt  login
root@cka-w2 ~/project-files (master)
$ git checkout bugfix/typo
Switched to branch 'bugfix/typo'
root@cka-w2 ~/project-files (bugfix/typo)
$ ls
app.txt  bufix.txt  login
root@cka-w2 ~/project-files (bugfix/typo)
$
```

git log --oneline --all

```
root@cka-w2 ~/project-files (bugfix/typo)
$ git log --oneline --all
d40c83f (master) style: improve footer spacing
3fc7beb (HEAD -> bugfix/typo) fix: correct footer typo
13b779a (feature/login) feat: implement login functionality
9aaeab0 feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (bugfix/typo)
$
```

```
root@cka-w2 ~/project-files (bugfix/typo)
$ git log --oneline --all
d40c83f (master) style: improve footer spacing
3fc7beb (HEAD -> bugfix/typo) fix: correct footer typo
13b779a (feature/login) feat: implement login functionality
9aaeab0 feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (bugfix/typo)
$ git checkout master
Switched to branch 'master'
root@cka-w2 ~/project-files (master)
$ git merge bugfix/typo -m "merge branch bugfix/typo"
Merge made by the 'ort' strategy.
 bufix.txt | 1 +
 1 file changed, 1 insertion(+)
 create mode 100644 bufix.txt
root@cka-w2 ~/project-files (master)
$
```

Merge commit b18b077

```
root@cka-w2 ~/project-files (master)
$ ls
app.txt  bufix.txt  layout.txt  login
root@cka-w2 ~/project-files (master)
$ git log  --onelie
fatal: unrecognized argument: --onelie
root@cka-w2 ~/project-files (master)
$ git log  --oneline
b18b077 (HEAD -> master) merge branch bugfix/typo
d40c83f style: improve footer spacing
3fc7beb (bugfix/typo) fix: correct footer typo
13b779a (feature/login) feat: implement login functionality
9aaeab0 feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (master)
$
```

git show → Merge: d40c83f 3fc7beb

```
fatal: unrecognized argument: --onelie
root@cka-w2 ~/project-files (master)
$ git log  --oneline
b18b077 (HEAD -> master) merge branch bugfix/typo
d40c83f style: improve footer spacing
3fc7beb (bugfix/typo) fix: correct footer typo
13b779a (feature/login) feat: implement login functionality
9aaeab0 feat: add footer section
fcaef20 style: improve application header styling
0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (master)
$ git show b18b077
commit b18b077f5bd1a2e7ce9e0f8d425e7e96a0ef6f72 (HEAD -> master)
Merge: d40c83f 3fc7beb
Author: vishal <vishalbhilare@test.com>
Date:   Tue Oct 13 11:39:59 2026 +0000

    merge branch bugfix/typo

root@cka-w2 ~/project-files (master)
$
```

45 / 45 git log --oneline --graph

```
root@cka-w2 ~/project-files (master)
$ git log --oneline --graph
*   b18b077 (HEAD -> master) merge branch bugfix/typo
|\
| * 3fc7beb (bugfix/typo) fix: correct footer typo
* | d40c83f style: improve footer spacing
|/
* 13b779a (feature/login) feat: implement login functionality
* 9aaeab0 feat: add footer section
* fcaef20 style: improve application header styling
* 0e3ffcd chore: initialize application
root@cka-w2 ~/project-files (master)
$
```